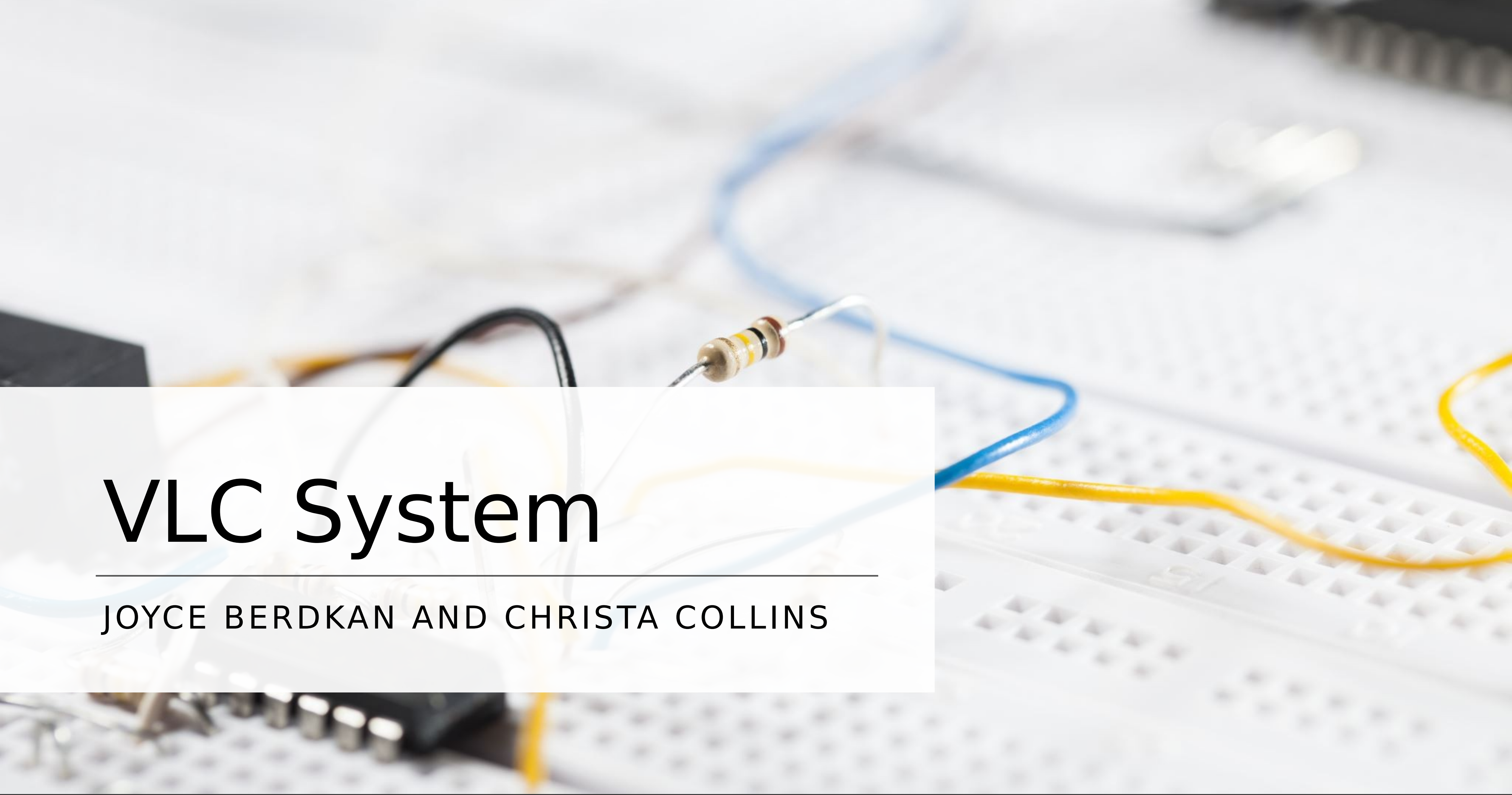


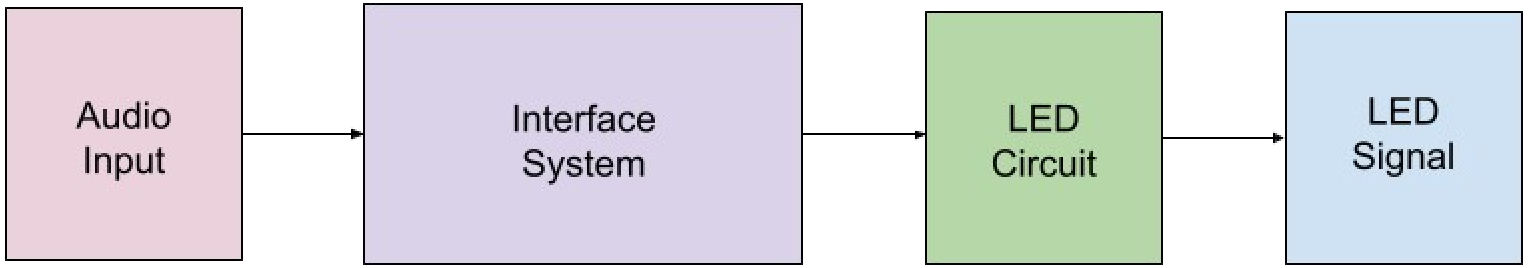
VLC System

JOYCE BERDKAN AND CHRISTA COLLINS

Project Objective

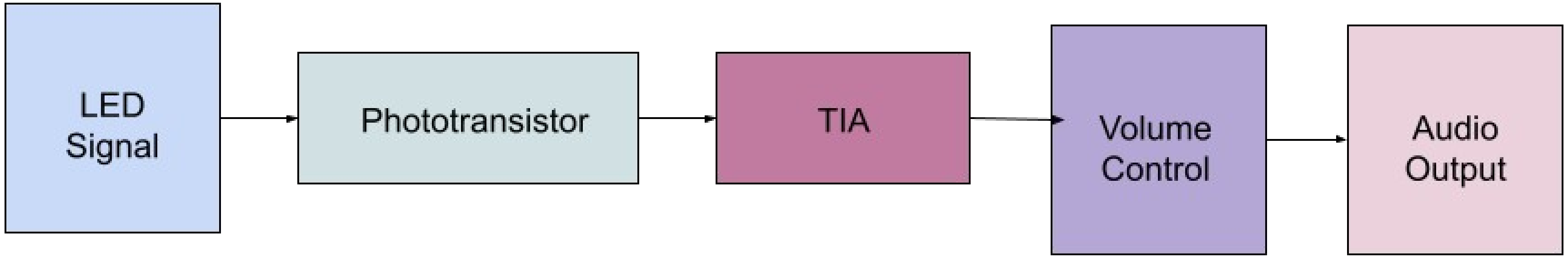
- This project's primary objective is to implement a communication system which both transmits and receives audio signals via visible light as a means of a carrier.
- This can be done by transmitting data signals through light.
- We accomplished this by transmitting audio signals through LEDs.

Block Diagram of Transmitter Circuit



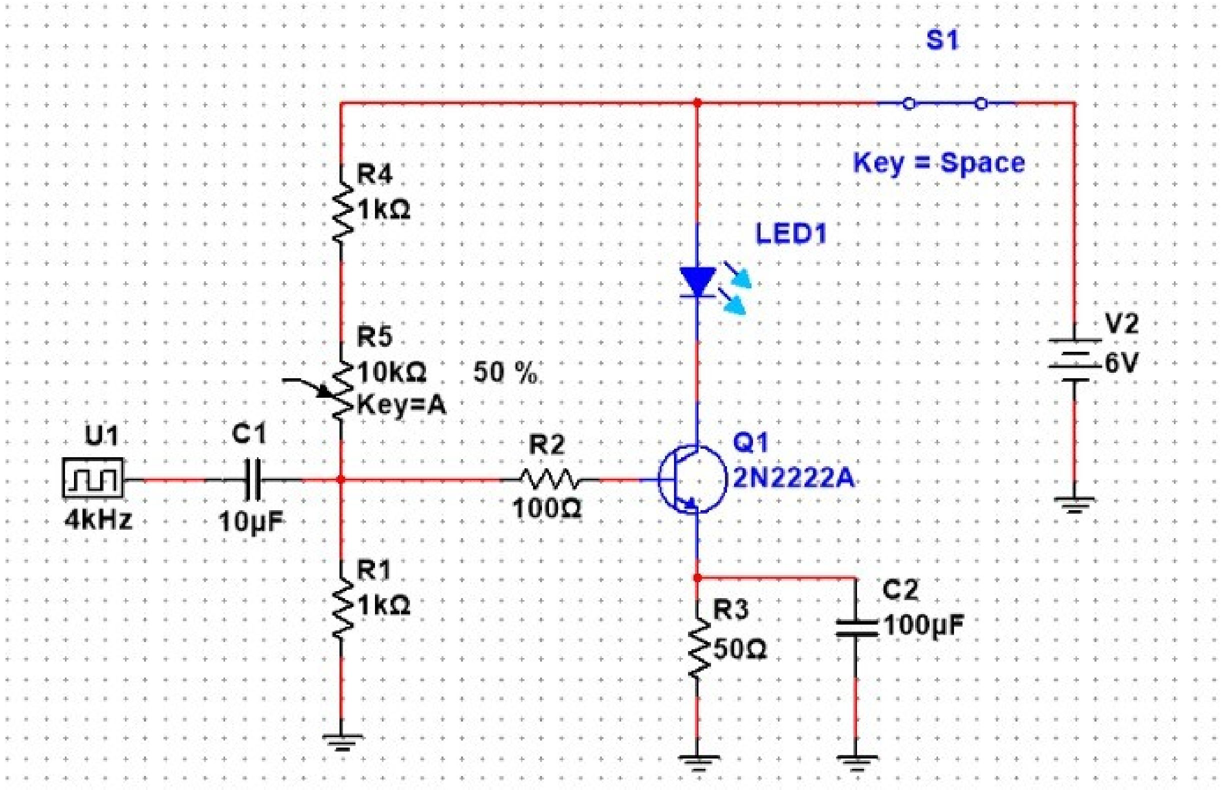
- Input audio is generated from an electric device
- Input goes through the circuit to drive the LED
- LED used to convert data into light pulses

Block Diagram of Receiver Circuit

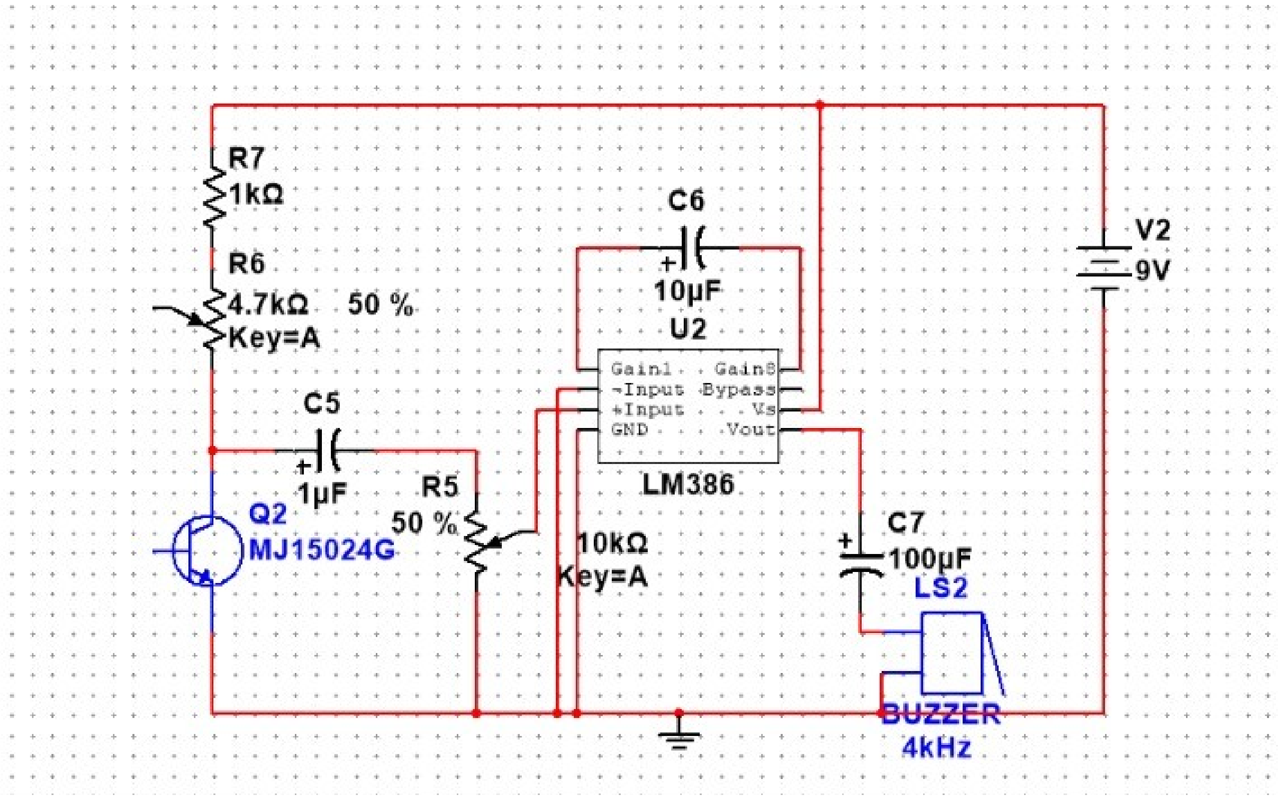


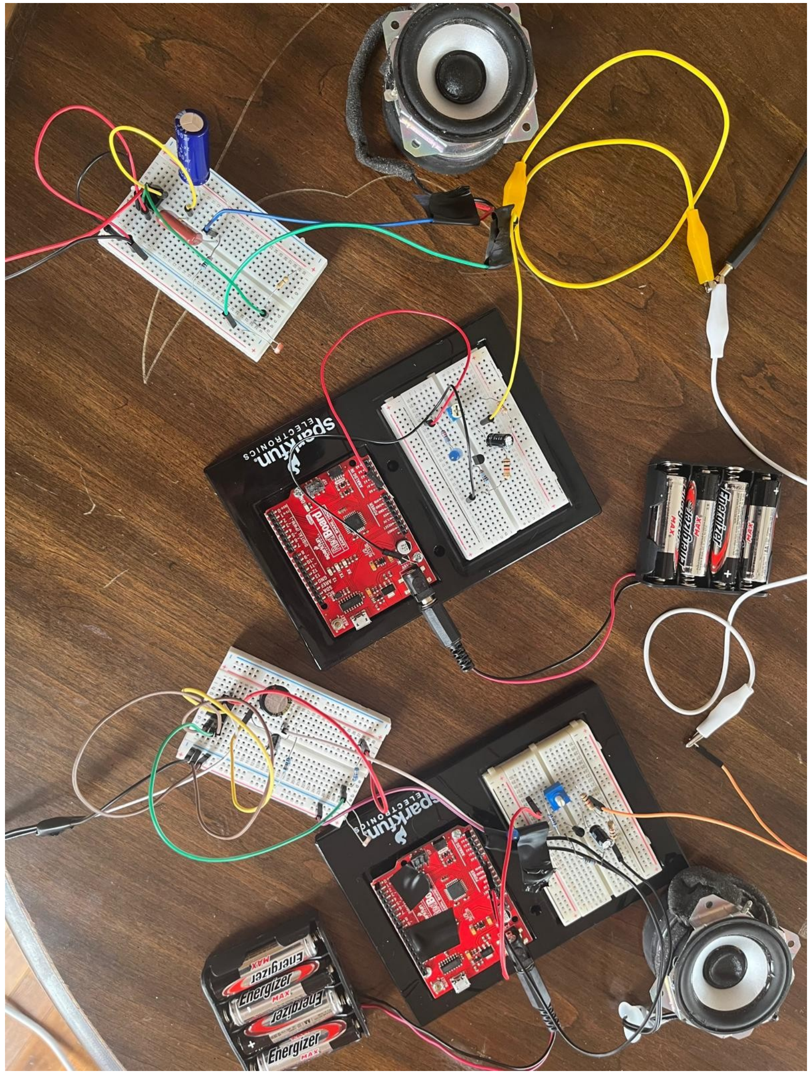
- Phototransistor collects light pulses and converts them to a current signal
- Current signal is converted into voltage
- Signal is then amplified for reception reliability

LED Driver [Transmitter] Circuit



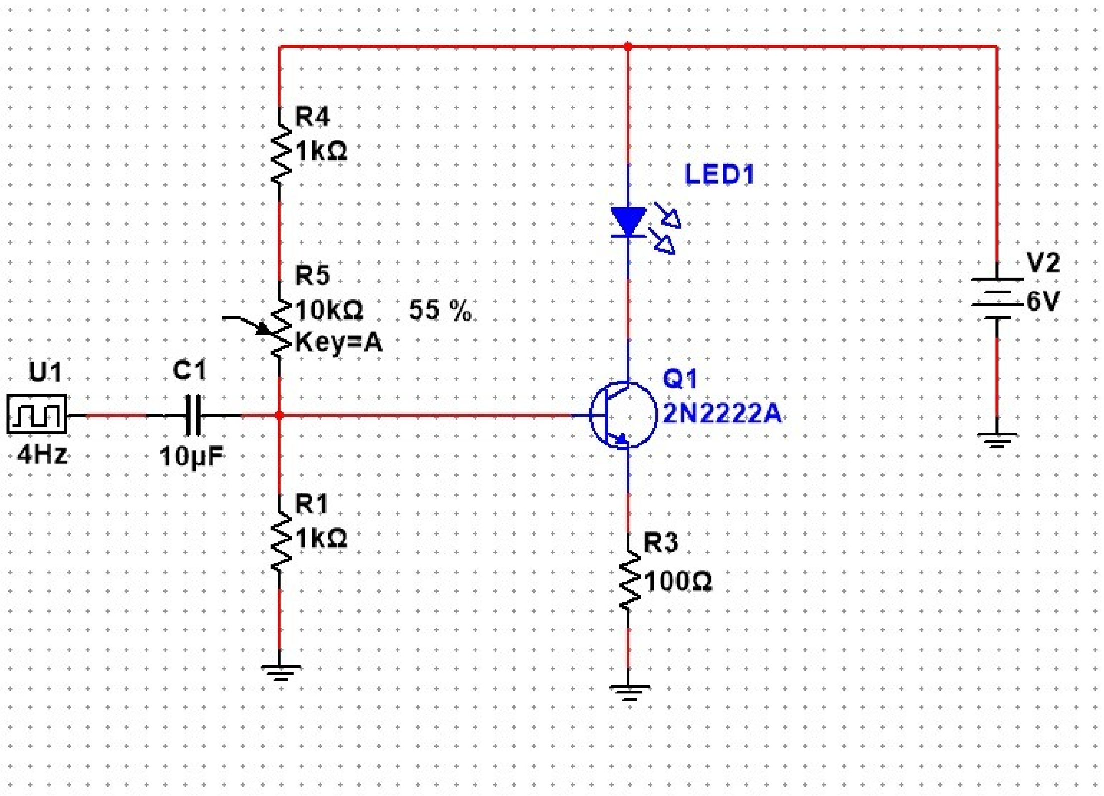
Photodiode [Receiver] Circuit



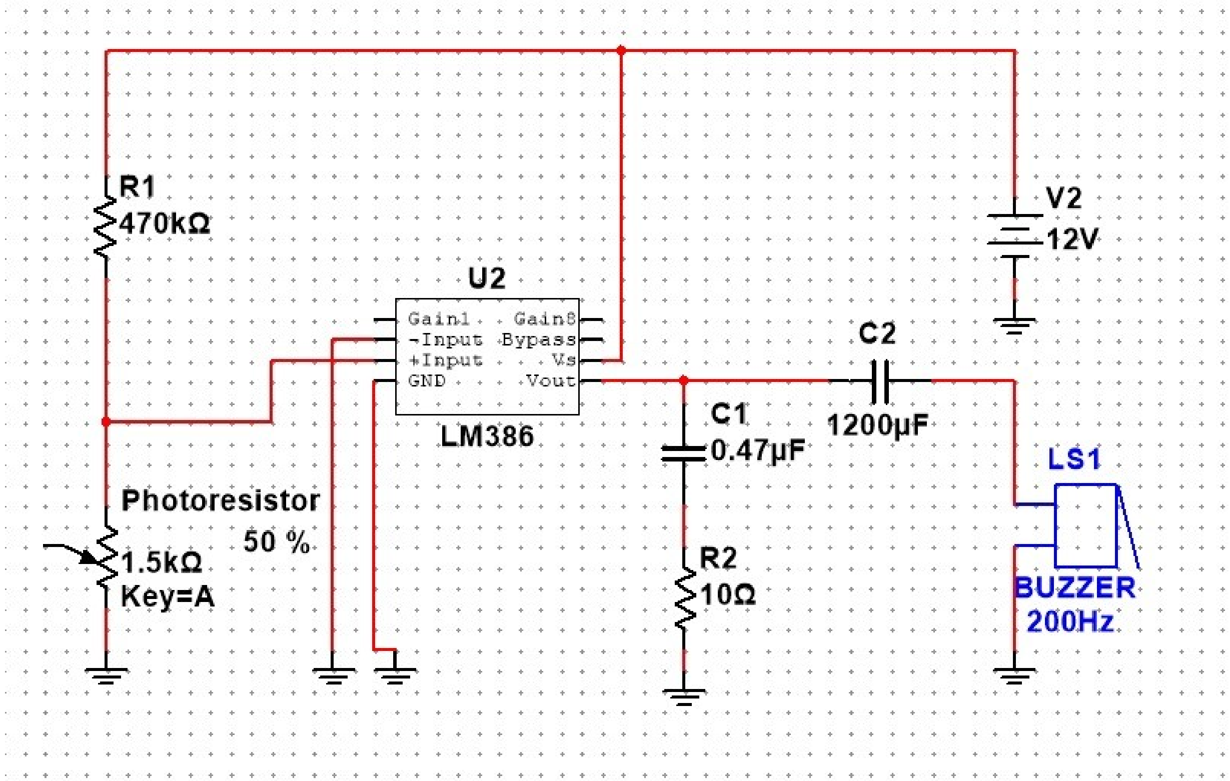


Final VLC System

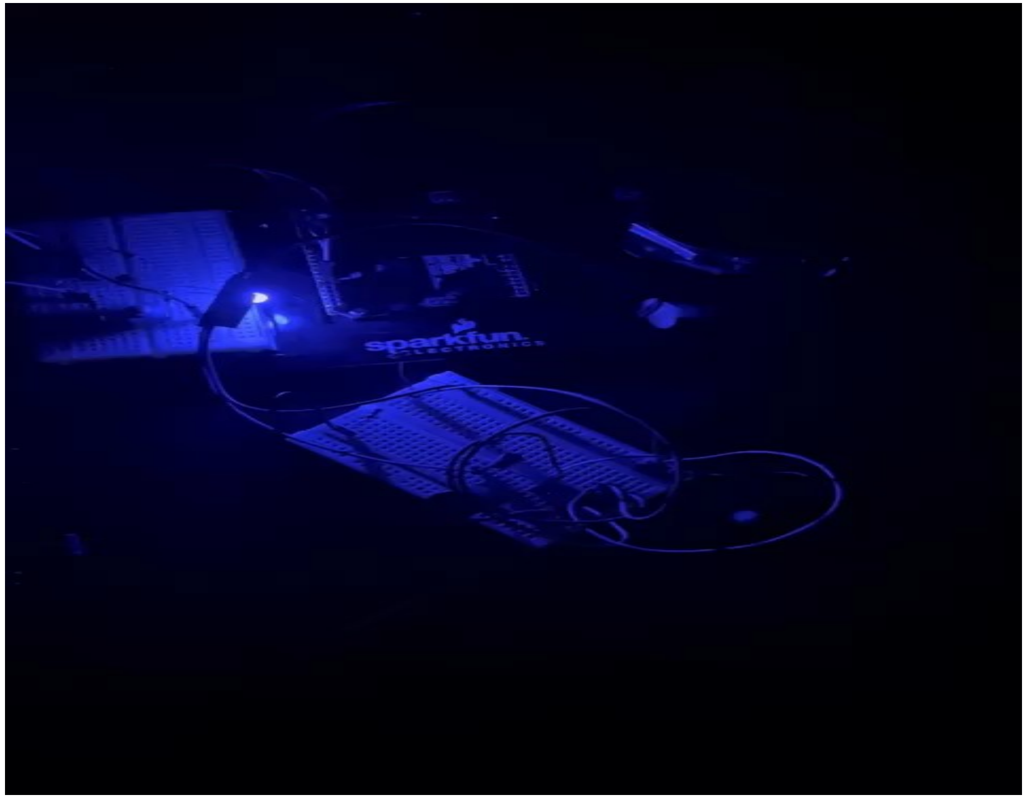
Modifications of Design [Transmitter]



Modifications of Design [Receiver]



Monophonic



Stereo

Results

- Left and right speakers played independently
- Cross talk between the left and right speaker channels due to light interference
- Range was not as good due to only having access to passive components instead of active components making the audio noisier

Discussion + Challenges

- Using photoresistors instead of phototransistors proved to be a challenge
- We had to use a voltage divider which made the audio noisier due to picking up the light signal using a DC supply which has a certain amount of ripple in it
- Photoresistor was extremely sensitive, audio sounded clear only at night
- The recommended capacitor sizes were not available, so the filtering quality was not as good
- We started out with a green LED but because it has a lower energy wavelength we switched to a blue LED for better range

Conclusion

- We built a stereo system from two monophonic systems which transmits audio signals over light waves and converts them back to audio signals